



FOREST VERIFICATION REPORT

ANALYZED ZONE INFORMATION

Latitude	-3.4653	Longitude	-62.2159
Zone size (km)	10	Analysis date	2026-09-29

80

FOREST INTEGRITY SCORE / 100 Healthy Forest

DETAILED RESULTS

Indicator	Value	Interpretation
Current vegetation (NDVI)	0.842	Excellent
NDVI Trend 3 years (Sentinel-2)	+0.025	Stable
10-year trend (Landsat 8/9)	+0.456	Improving
40-year trend (Landsat 1984)	+0.493	Improving
Forest Status (FAO)	Forest recovery	Recuperation
Annual NDVI rate	+1.23% / year	Amelioration
NDVI change 5 years	+0.007	Stable
Leaf type	Mixed (Coniferous/Broadleaf)	Mixed needles and flat leaves
Developmental stage	Old-growth forest (>120 years)	Ancient forest, high biodiversity
Above-ground biomass	~78.4 t/ha	Medium biomass (50-150 t/ha)
Canopy Height (GEDI+S2 10m)	13.2m max / 7.3m median	Tree height from LiDAR (ISS) fused with Sentinel-2 at 10m resolution
Carbon Assessment (IPCC)	1,337,536 t CO ₂	CO ₂ sequestered in this forest zone — IPCC methodology (AGB+BGB × 0.47 × 3.67)
CO ₂ per hectare	170.3 t CO ₂ /ha	Carbon density per hectare
Zone area	7,854 ha	Analysis zone area in hectares
Carbon credit value	EUR 20,063,040	Estimated value at EUR 15/tonne CO ₂ (voluntary carbon market)
REDD+ MRV Monitoring	Compliant	This report provides satellite-based Measurement, Reporting and Verification (MRV) data for the analyzed forest zone, consistent with REDD+ monitoring requirements. CO ₂ sequestration calculated using IPCC Tier 1 methodology from NASA GEDI LiDAR biomass data.

Methodology	IPCC 2006/2019 — Tier 1	Above-ground biomass measured via NASA GEDI LiDAR (25m, ISS). Carbon fraction: 0.47. CO2 conversion factor: 3.67. Below-ground biomass ratio: 0.26 (IPCC root-to-shoot). Source: IPCC Guidelines for National GHG Inventories (2019).
Carbon credit potential	EUR 1,671,920 / year	Estimated annual value based on maintained forest preservation. For official REDD+ credit issuance under VCS or Gold Standard, independent third-party verification is required.
GEDi Tree Cover	100%	Percentage of the zone covered by trees according to NASA GEDI
AI Forest Type	Dense Tropical Forest	High biodiversity, dense canopy
Sentinel-1 Radar (VV)	-7.805 dB	Dense forest (radar)
Forest cover	99.9%	Excellent
Fire pixels detected (FIRMS/year)	146 pixels	~36 events estimated
Overall trend	stable	
Satellites used	- ESA Sentinel-2 - NASA FIRMS - ESA Sentinel-1 - NASA Landsat 8/9 - NASA Landsat 4/5 - NASA GEDI LiDAR	
Deforestation Alert	STABLE — No significant change	
Area change (3 years)	+3 ha	+0.03%
Annual rate (area/yr)	+1 ha/yr	Recovery
Area change (10 years)	+9060.5 ha	+988.7%
Annual rate 10yr (area)	+906 ha/yr	Recovery

SCORE BREAKDOWN

Component	Points	Maximum
Current vegetation (NDVI)	25	25
NDVI trend (3 years)	15	25
Forest cover	25	25
Sentinel-1 Radar (VV)	15	15
Fire score (0=many fires, 10=none)	0	10
TOTAL	80	100

SPECIES CLIMATE SUITABILITY (2050)

Scenario: SSP2-4.5 (2050) | Temperature: 26.6°C → 30.1°C (+3.5°C) | Precipitation: 2612 → 1990 mm/year (-23.8%)

Species	Family	Score 2050	Status	Temp. range
Maclobium acaciifolium	Fabaceae	0.72	Suitable	13.1–29.0°C
Aechmea fernandae	Bromeliaceae	0.71	Suitable	19.3–28.8°C
Isertia hypoleuca	Rubiaceae	0.64	Moderate risk	18.8–28.9°C
Couratari tenuicarpa	Lecythidaceae	0.62	Moderate risk	19.4–28.5°C
Inga pilosula	Fabaceae	0.56	Moderate risk	12.5–28.5°C
Victoria amazonica	Nymphaeaceae	0.54	Moderate risk	19.8–27.8°C
Mussaenda erythrophylla	Rubiaceae	0.53	Moderate risk	10.7–28.8°C
Hirtella racemosa	Chrysobalanaceae	0.51	Moderate risk	14.0–28.5°C
Eleocharis filiculmis	Cyperaceae	0.5	Moderate risk	13.4–28.5°C
Pistia stratiotes	Araceae	0.49	Moderate risk	-1.6–28.7°C
Pontederia crassipes	Pontederiaceae	0.47	Moderate risk	4.0–28.6°C
Neptunia prostrata	Fabaceae	0.45	Moderate risk	12.1–28.3°C
Couepia paraensis	Chrysobalanaceae	0.45	Moderate risk	16.9–27.7°C
Cyperus haspan	Cyperaceae	0.4	Moderate risk	7.3–28.1°C
Triplaris gardneriana	Polygonaceae	0.32	At risk	13.1–27.1°C

FIRE DETECTION METHODOLOGY: Fire pixels are detected by NASA FIRMS using VIIRS (375m) and MODIS (1km) thermal anomaly sensors. Raw pixel count represents thermal detections over 12 months in the analysis zone. Estimated fire events are calculated using NASA clustering methodology (~4 pixels per fire event average). This is a statistical approximation. For precise fire event mapping, consult NASA FIRMS directly at firms.modaps.eosdis.nasa.gov

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